

Predicting the Future Performance of the Planned Seismic Network in Chinese Mainland

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Abstract

The China Earthquake Administration has currently launched an ambitious nationwide seismicity monitoring network project that will increase the number of stations from ~ 950 to 2000 for the broadband seismic stations used to compile the earthquake catalog. The new network is planned to go online by the end of 2023. For more than half of Chinese mainland, the interstation distance of the broadband seismic network will soon be smaller than 100 km, for 27% smaller than 50 km, and for 6% smaller than 25 km. Of all possible ways to characterize the higher-resolution monitoring of the frequent smaller earthquakes expected inside Chinese mainland, the completeness magnitude (M_c) remains one of the most commonly used. Using the prior model of the Bayesian magnitude of completeness method calibrated on the Chinese earthquake catalog from 1 January 2009 to 26 June 2022, we predict the spatial distribution of M_c for the new network based on the planned network configuration. If almost the entire Chinese mainland is at present covered down to $M_c = 3.3$, this threshold will fall to $M_c = 2.9$ in the near future. This means approximately two times more earthquakes will be recorded in the complete catalog available for statistical analysis per year (for $a = 6.77$ and $b = 0.80$ in the Gutenberg–Richter law $\log_{10} N = a - b \cdot M$, in which N represents the number of events of magnitude larger than or equal to M and $M \geq M_c$). Based on the observation that abnormal seismicity as precursors are most likely to be observed at least at three units below the mainshock magnitude, and assuming earthquakes to be potentially damaging at $M \geq 5$, the new seismic network shall achieve the goal of 76% coverage for optimal seismic-based earthquake prediction research.

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[Supplemental Material](#)

Introduction

The densification of a seismic network improves earthquake detectability to smaller magnitudes (Engdahl and Bondár, 2011). In turn, access to smaller earthquakes improves statistical seismology studies by providing more data and thus better insights into the underlying crustal processes. Interestingly, microseismicity has been shown to be critical for the observation of seismic precursors prior to large and potentially damaging earthquakes (Jones *et al.*, 1982; Bernard *et al.*, 1997; Wang *et al.*, 2006; Ebel, 2008; Mignan, 2014; Brodsky, 2019; Trugman and Ross, 2019). Mignan (2014) proposed the following empirical law based on a meta-analysis of earthquake precursor studies:

$$m_{\min} < M - 3.0, \quad (1)$$

in which $m_{\min}$ is the minimum foreshock magnitude threshold for seismic events prior to a mainshock of magnitude M that is necessary to start distinguishing seismic precursors from the

normal behavior of seismicity. The completeness magnitude M_c is the simplest, and one of the best proxies to the detection capability of a seismic network. Earthquake data, considered complete above this limit, follow the Gutenberg–Richter frequency–magnitude distribution (FMD). The optimal use of data requires that $m_{\min} = M_c$. As of 2011, the minimum M_c covering almost the entire Chinese mainland was about 3.7 (Mignan *et al.*, 2013).

Chinese mainland is a highly active seismic zone, as depicted in Figure S1, available in the supplemental material

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to this article. The seismic network in this region plays a crucial role in the global seismicity database, offering valuable insights into internal plate tectonic activity, Earth's interior structure (e.g., [Li et al., 2008](#); [Han et al., 2022](#)), seismicity in neighboring regions, and national and global earthquake surveillance. The establishment of numerous monitoring stations in Chinese mainland has resulted in significant benefits for enhancing earthquake prediction and advancing seismic research (e.g., [Mignan et al., 2013](#); [Wu, 2018](#); [Wu et al., 2019](#); [Zhang et al., 2021](#)). Over the past decade, the detection capability, or M_c , of the seismic network in Chinese mainland has been improved with the upgrading of processing techniques (e.g., [Wang et al., 2017](#)), whereas the basic network layout remains unchanged. The China Earthquake Administration (CEA) has recently launched an ambitious nationwide network construction project that aims to extend the scale and applications (e.g., earthquake early warning; [Li, 2023](#)) of the seismic network. It involves establishing around 15,000 stations with an investment of ~ 1.87 billion Renminbi (RMB; \\$290 million U.S.; see [Data and Resources](#)). The planned nationwide seismic network comprises three interconnected networks: the seismicity monitoring network with 2000 datum stations, the strong-motion network with 3000 basic stations, and the low-cost intensity network with 10,000 microelectromechanical system (MEMS) sensors. All these networks are integrated into an earthquake intensity reporting and early warning system. However, only the datum stations in the seismicity monitoring network will be used for catalog compilation. This project will increase the number of stations available for catalog compilation from around 950 to 2000 in the future (Fig. 1), with real-time data transmission being planned. The customary assessment methods (e.g., [Mignan and Woessner, 2012](#)) to estimate the M_c are hampered for this network at the planning, design, and implementation stages by a lack of observations.

The Bayesian magnitude of completeness (BMC) method proposed by [Mignan et al. \(2011\)](#) provides a promising solution to address the above problem of lack of observations to estimate M_c using Bayesian inference relating the spatial density of stations in the network to M_c values, weighted by the respective uncertainties of the observations and the prior model. This method has been successfully applied in observed induced ([Mignan, 2021](#)) and natural seismicity, for example, Taiwan region ([Mignan et al., 2011](#); [Feng et al., 2022](#)), Chinese mainland ([Mignan et al., 2013](#)), Switzerland ([Kraft et al., 2013](#)), Lesser Antilles arc ([Vorobieva et al., 2013](#)), California ([Tormann et al., 2014](#)), Greece ([Mignan and Choularas, 2014](#)), Iceland ([Panzera et al., 2017](#)), South Africa ([Brandt, 2019](#)), and Venezuela (see [Data and Resources](#)).

As an updatation of the analysis applying BMC method to Chinese mainland by [Mignan et al. \(2013\)](#), we begin this article with an evaluation of detectability capability based on the earthquake catalog accumulated from 2009 to 2022 inside Chinese mainland. Once the BMC method is calibrated for

the existing seismic network, we then extend its application to predict the future performance of the planned network. Finally, we quantify the potential improvement of seismicity in a more complete catalog observed with the change in the network and investigate the significance for seismic-based earthquake prediction research based on the optimal use of data, namely $m_{\min} = M_c$. Our work will serve as an important reference to guide the design and optimization of the planned seismic network upgrade in Chinese mainland.

Data

Earthquake catalog (2009–2022)

For our purpose, we define the inside of Chinese mainland as the region within 100 km outside of the boundaries of Chinese mainland and Hainan Island (~ 11.5 million km²), where the China Earthquake Networks Center (CENC) produces a catalog based on the same standard and workflow from 2009 when the existing seismic network became operational. Approximately 1,128,068 earthquakes, in which 23,693 with $M_L \geq 3.0$, that occurred inside Chinese mainland were reported in the catalog provided by the CENC from 1 January 2009 to 26 June 2022 (Fig. S1). Among them, the largest and the most damaging earthquake was the 2008 M_s 8.0 Wenchuan earthquake, which caused $\sim 70,000$ deaths and 20,000 missing people.

The seismic networks

The existing seismic network in Chinese mainland that was used to compile the catalog by the CENC consists of ~ 950 high-gain broadband seismic stations (Fig. 1a). The new network is being constructed as a part of the National System for Fast Report of Intensities and Earthquake Early Warning project of China. This project, led by the CEA, was launched and implemented in 2015 and 2018, respectively. The new network consists of a total of ~ 2000 datum stations equipped with three-component broadband seismometers and accelerometers (Fig. 1b). The majority of existing stations will be upgraded and incorporated into the new network, whereas a few stations will be excluded due to unfavorable site conditions and high ambient noise (Yang Zang, written comm., 2023). It is expected that, by end of 2023, the new network will officially be online (Wenhui Huang and Chaoyong Peng, written comm., 2021).

We computed the interstation distance from the average distance of a given site to its three closest broadband stations ([Kuyuk and Allen, 2013](#); [Li and Wu, 2016](#); [Li et al., 2021](#)) using a mesh of $0.1^\circ \times 0.1^\circ$ resolution. In half of Chinese mainland, the interstation distances of the existing network are less than 100 km, in 14% less than 50 km, and in 0.8% less than 25 km (Fig. S2a). In contrast, in 57% of Chinese mainland, the interstation distance will soon be smaller than 100 km, in 27% smaller than 50 km, and in 6% smaller than 25 km (Fig. S2b). Although the area with an interstation distance smaller than 100 km to be covered by the new network is almost the same

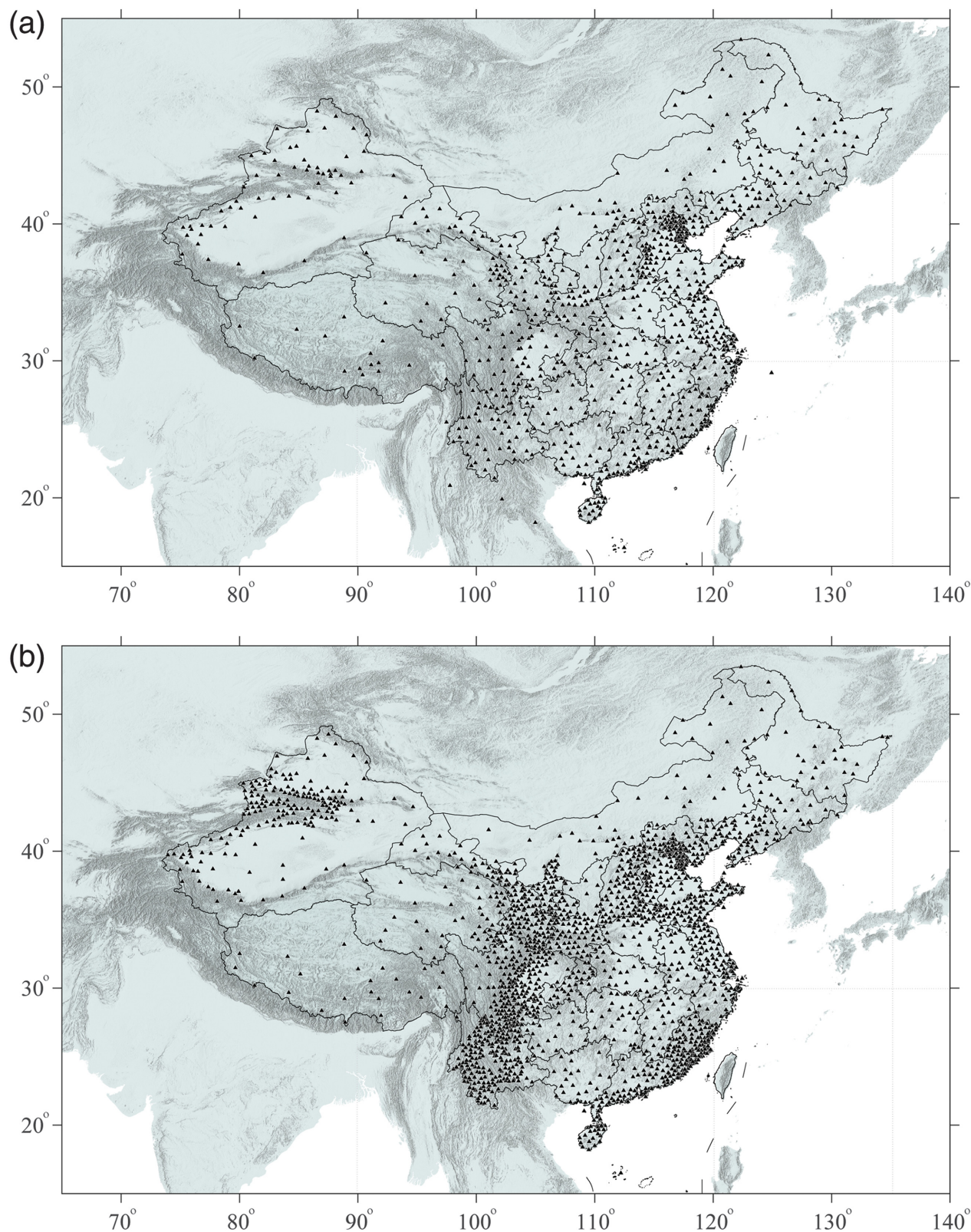
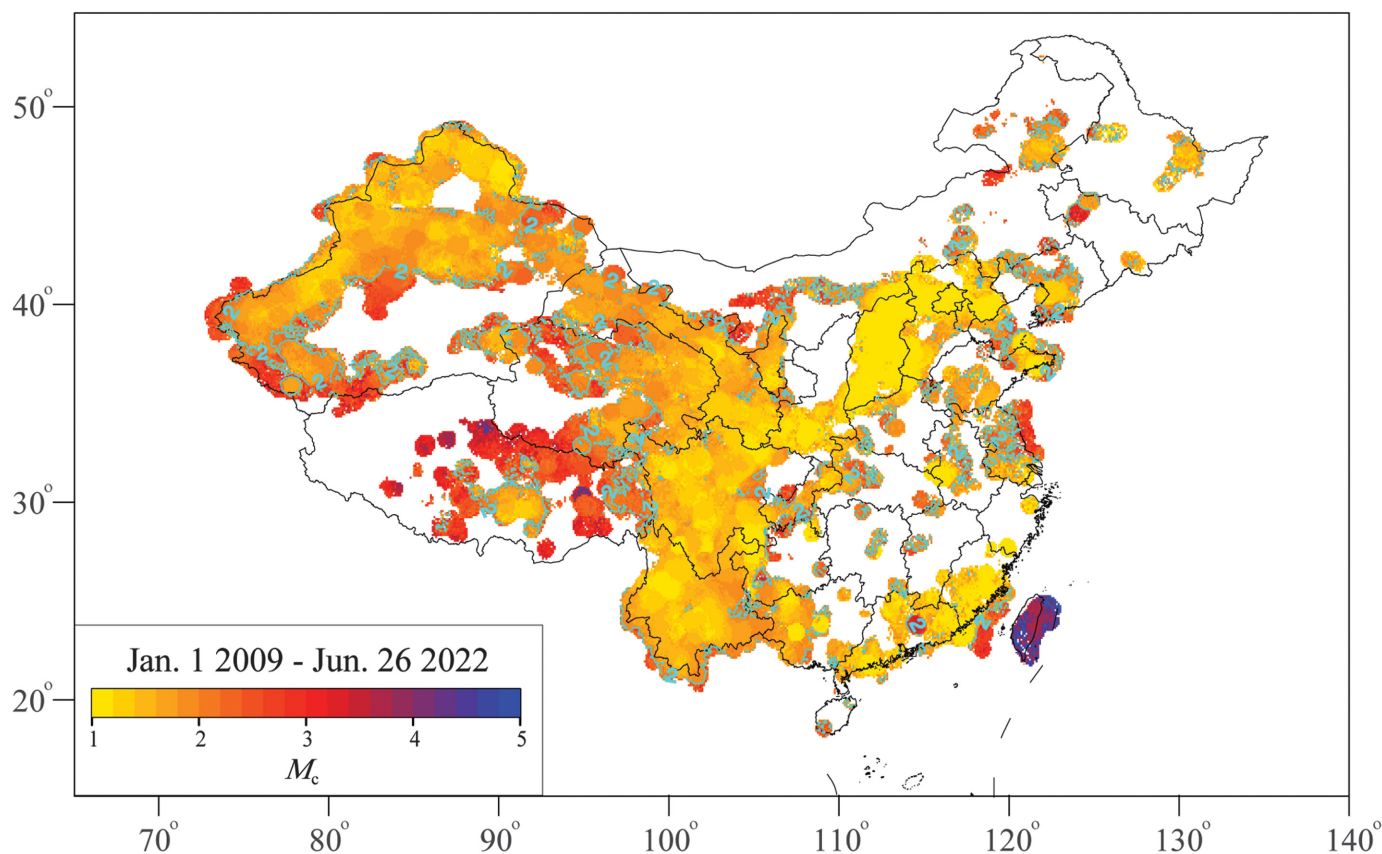


Figure 1. (a) Existing (as of 2018) ~950 broadband seismic stations (black triangles) as part of the China Earthquake Networks Center (CENC). (b) Planned (as of 2023) ~2000 stations (black triangles) to be newly deployed or upgraded in the National

System for Fast Report of Intensities and Earthquake Early Warning of China. We define “inside” in this study as the region within 100 km outside of Chinese mainland’s boundaries.



as that covered by the existing network, the areas with an inter-station distance smaller than 50 km and 25 km are expanded by factors of 2 and 8, respectively. The density of the new network is obviously improved. The new network will have an interstation distance of $\sim 20\text{--}30$ km in North China, the central China north–south seismic belt, the southeast coast, the middle section of the Tianshan Mountains in Xinjiang (Fig. S2b). Overall, the density of the new network in North China, the central China north–south seismic belt, the southeast coast, the middle section of the Tianshan Mountains in Xinjiang will be comparable to the most dense seismic network worldwide, for example, Japan, the U.S. west coast, and Taiwan regions (e.g., Kuyuk and Allen, 2013).

Monitoring Capability Estimated by M_c Past network performance

Figure 2 shows M_c maps for time periods from 2009 to 2022. We use the FMD-based mapping method (Wiemer and Wyss, 2000) with grid resolution of $0.1^\circ \times 0.1^\circ$ for each of them, and define a cylinder with a fixed radius of $r = 50$ km and the minimum number of earthquakes $N_{\min} = 50$. The bulk M_c in each grid is computed based on the events within a cylinder using the nonparametric median-based analysis of the segment slope (MBASS) technique (Amorèse, 2007), and its deviation shown in Figure S3 is obtained from 200 bootstrap samples (Efron, 1979). Many gaps remain in grid points with low seismicity within the cylinder. Simply increasing r is not recommended

Figure 2. Map of the observed M_c for the time period from 2009 to 2022. The median M_c based on the M_c spatial distribution is 1.7. The M_c estimates are computed using the median-based analysis of the segment slope (MBASS; Amorèse, 2007) and based on the frequency–magnitude–distribution-based mapping method (Wiemer and Wyss, 2000). Map of the standard deviation of the observed M_c is shown in Figure S3.

due to possible oversmoothing with obvious artifact patterns (Mignan and Woessner, 2012). The BMC technique is based on the density of the seismic network for determining r (Mignan et al., 2011). This method efficiently decreases the number of gaps, but it can be used only where network information is available.

The overall spatial M_c between 1.5 and 2.0 inside Chinese mainland in Figure 2 is basically the same as that from 1 October 2008 to 31 August 2011 of Mignan et al. (2013), which used grid resolution of $0.5^\circ \times 0.5^\circ$ and estimated a median M_c value of 1.6. Thanks to the improved spatial resolution and 10 more years of seismicity accumulation, Figure 2 shows reduced gaps and more details of the spatial distribution of M_c , compared with the results of Mignan et al. (2013). The median M_c values based on its spatial distribution in Figure 2 are 1.7. Figure 2 produced by the mapping method also exhibits many gaps in regions with low seismicity where M_c cannot be computed with confidence. The M_c values outside Chinese mainland (e.g., Taiwan region) are larger, because earthquakes were

located by the CENC using only the national broadband seismic network in this case. The areas with M_c values smaller than 1.5 account for 17% of the total inside Chinese mainland, and values smaller than 2.0 correspond to 37%. Areas with M_c values smaller than 1.5 are sparse in the central China north-south seismic belt, northern part of Xinjiang, North China, Southeast China, and Lhasa.

Previous studies show that M_c has a negative correlation with the network density (Nanjo *et al.*, 2010; Mignan *et al.*, 2011), which is quantified via the interstation distance in this study. We observe that M_c is strongly heterogeneous (Fig. 2) due to the nonuniform spatial distribution of stations (Fig. 1a). The site conditions of stations, namely ambient noise, are not considered when we estimate M_c , because the impact on M_c from factors beyond network density does not vary in time and space, which are both validated by Nanjo *et al.* (2010) and Figure 2 in this study accompanying the results of site condition depicted in figure 5 of Chen *et al.* (2020), respectively. Stations of the existing network in Chinese mainland have an ambient noise level smaller than -140 dB. The majority of them are bedrock stations, and the rest are borehole stations (Yang Zang, written comm., 2023), which is the potential reason for the fact that the ambient noise has an insignificant effect on M_c in space in Chinese mainland.

BMC application

The BMC method proposed by Mignan *et al.* (2011) merges the observed M_c with prior Bayesian information, which is deduced from the relationship between the density of the seismic network and M_c observations, weighted by their respective standard deviations. The posterior M_c^{post} is obtained in a two-step procedure. First, a spatial resolution optimization is implemented. However, we use the mapping method with fixed parameters to obtain the spatial distribution of M_c to avoid the iterated use of the network information in observed M_c and prior Bayesian information, at the cost of the nonoptimization of the spatial resolution (in contrast to previous BMC applications in other regions). Therefore, only the FMD-based mapping method is adopted for grids with smaller spatial resolution ($0.1^\circ \times 0.1^\circ$) in this article to produce Figure 2. In addition, we still use the MBASS technique instead of the maximum curvature (MAXC) approach to consider the possible M_c heterogeneities introduced by the aforementioned choice. The MAXC approach is a standard method of estimating the observed M_c in the BMC method while resulting in large uncertainty, for example, $\sigma = 0.56$ in Mignan *et al.* (2013) for the Chinese catalog. The MBASS technique is here shown to reduce those uncertainties in the context of this study. Second, a Bayesian approach to merge the observed M_c^{obs} and the prior M_c^{pred} is completed using Gaussian conjugates, giving the posterior M_c^{post} for each grid as

$$M_c^{\text{post}} = \frac{M_c^{\text{pred}} \sigma_0^2 + M_c^{\text{obs}} \sigma^2}{\sigma_0^2 + \sigma^2}, \quad (2)$$

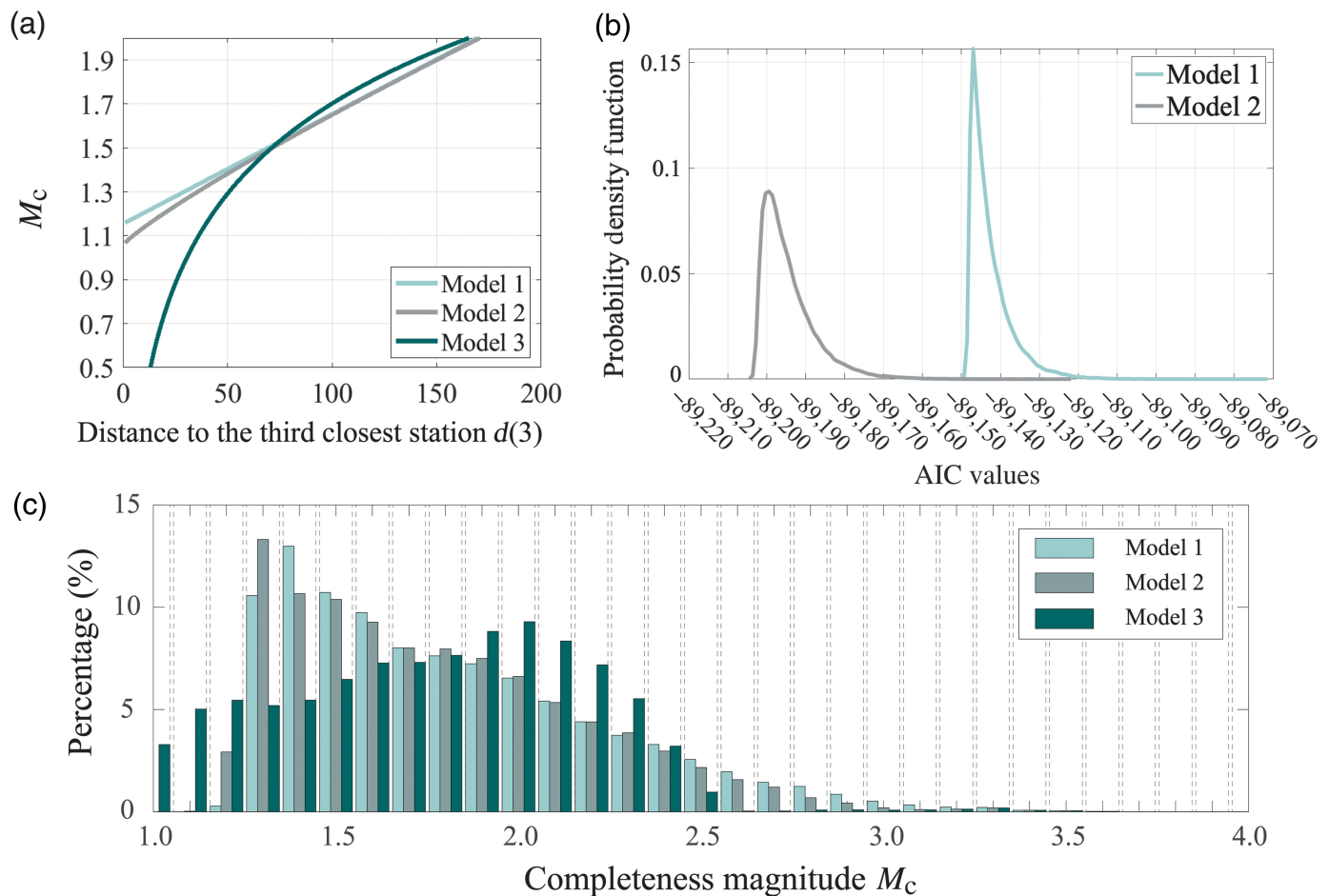
in which σ and σ_0 are the standard deviations for M_c^{pred} and M_c^{obs} , respectively. The posterior standard deviation σ_{post} can be estimated as

$$\sigma_{\text{post}} = \sqrt{\frac{\sigma_0^2 \sigma^2}{\sigma_0^2 + \sigma^2}}. \quad (3)$$

For the prior model, $M_c^{\text{pred}} = f[d(3)]$, in which $d(3)$ is the distance to the third closest station that is used to fit the observed M_c . The CENC generally compiles catalog by using waveforms from at least three stations (Ce Jiang, written comm., 2022). As a novelty, we compare three models: model 1 $M_c^{\text{pred}} = ad(3) + b$, with a degree-of-freedom (df) of 2; model 2 $M_c^{\text{pred}} = ad(3)^c + b$, with a df of 3; and model 3 $M_c^{\text{pred}} = a \log_{10} d(3) + b$, with a df of 2. Figure 3a demonstrates that model 3, with its logarithmic-type function, predicts a more pronounced decrease in M_c values with network density, followed by model 2 and model 1. Figure 3b shows the binned distributions of M_c values predicted by the models for the planned network inside Chinese mainland, revealing notable disparities between model 1 and model 2 when $M_c \leq 1.4$, with model 2 predicting smaller M_c values. The observed grid value M_c from 2009 to 2022 (Fig. 2) is used to evaluate the alternative models. The four indicators are the sum of squares due to error (SSE), the coefficient of determination (R -square), the root mean squared error (rmse), and the Akaike information criterion (AIC; Akaike, 1970). We found that model 2 $M_c^{\text{pred}} = ad(3)^c + b$ performs best in terms of both the goodness of fit (SSE = 12,659, R -square = 0.3714, and rmse = 0.4657) and model evaluation (AIC = $-89,203$; Table 1). This model is exactly the one used in the former BMC, for example, for Taiwan region (Mignan *et al.*, 2011).

To assess the contrast in AIC values between models 1 and 2, we employ the concept of relative likelihood according to Burnham and Anderson (2002). Let us define $\text{AIC}_{\min}$ as representing the smallest AIC value of the two models, which in our case corresponds to model 2. The relative likelihood, calculated as $\exp[(\text{AIC}_{\min} - \text{AIC}_{\text{model1}})/2]$, quantifies the probability that model 1 could minimize the estimated information loss compared to model 2. Our calculations reveal a relative likelihood of 3.8×10^{-11} for model 1 compared to model 2, indicating that model 1 is significantly less probable than model 2 in terms of minimizing information loss. This relative likelihood indicates the probability of model 1 being the true model, given the data and the referenced models. Essentially, the value of 3.8×10^{-11} for model 1 signifies that model 2 is a significantly superior candidate for the Kullback–Leibler best approximation to the true model, given the available data.

To further analyze the contrast in AIC values between the two models, we generate 50,000 Monte Carlo random samples for M_c at each grid using a normal distribution $N(M_c^{\text{obs}}, \sigma_0^2)$, in which M_c^{obs} and σ_0 are obtained from Figure 2 and Figure S3, respectively. We fit model 1 and model 2 to these samples and



calculate 50,000 AIC values. Using these values, we construct the probability density function (PDF) of AIC for each model. The PDFs of AIC for model 1 and model 2 replace the point estimates of $-89,155$ (model 1) and $-89,203$ (model 2), enabling us to assess the differences between them. Figure 3c confirms that the two PDFs are clearly separated, indicating that model 2 outperforms model 1 significantly. For example, the probability of model 1 producing an AIC value equal to or smaller than $-89,203$ is 0, whereas the probability of model 2 producing an AIC value equal to or higher than $-89,155$ is 0.0011. This substantial disparity between the two distributions firmly establishes model 2 as the clear winner with the smallest AIC value.

Figure 3. (a) Plot of the functional dependence M_c of the models given in Table 1 as a function of $d(3)$ —the distance to the third closest station. (b) Probability density functions (PDFs) of the Akaike information criterion (AIC) values for model 1 and model 2, respectively. (c) Binned distributions of M_c values predicted by the three models for the planned network inside Chinese mainland.

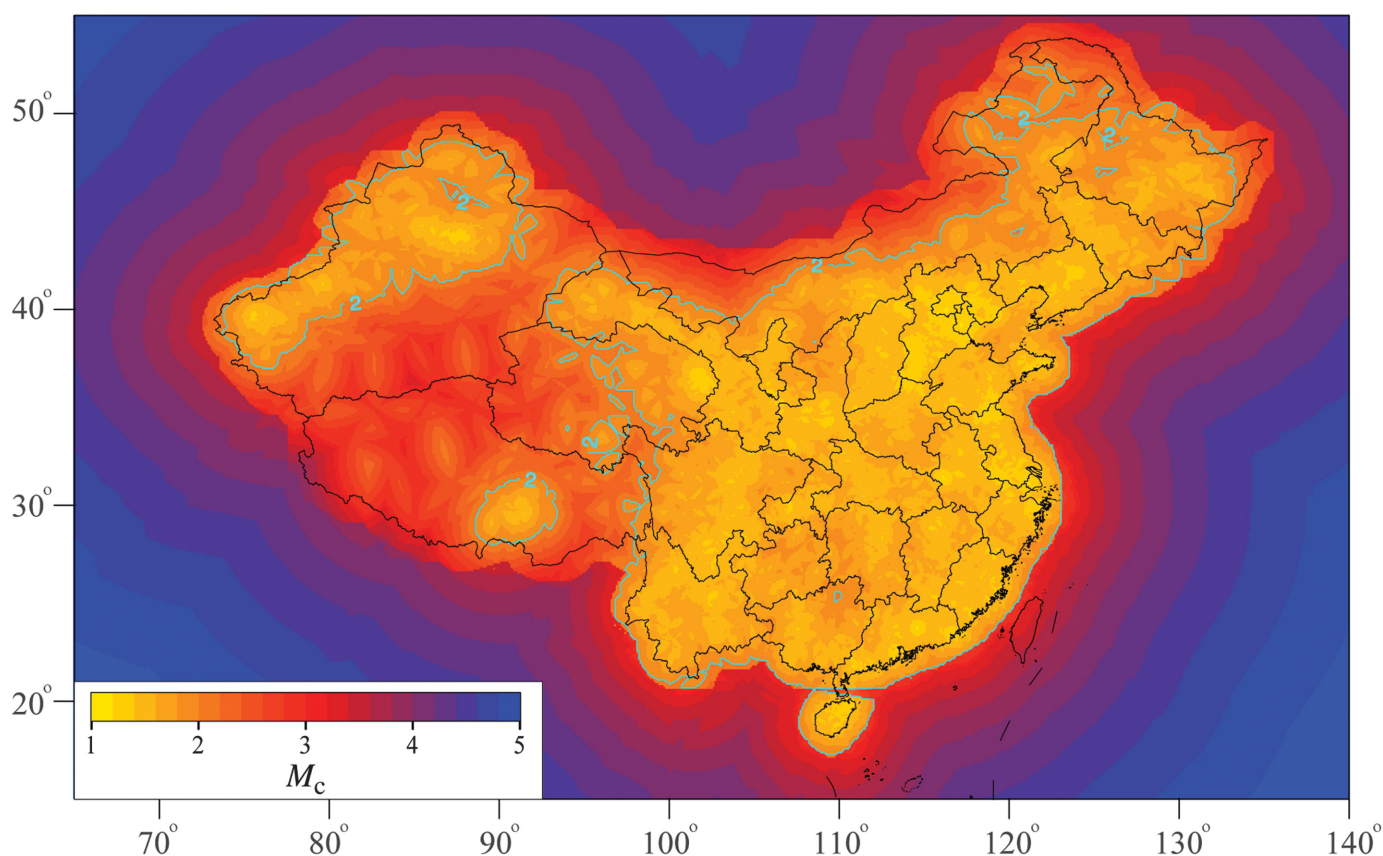
The physical basis of model 2, grounded in seismology (Mignan *et al.*, 2011), takes into account the relationship between earthquake magnitude and the dominant frequency and maximum amplitude peak of recorded seismograms used to measure M_L . Smaller events tend to have higher dominant frequencies and, as higher frequencies attenuate more rapidly,

TABLE 1

Characterization Parameters of Alternative Models and the Results of Their Evaluation to Observed M_c Values

Numbers	Model	<i>a</i>	<i>b</i>	<i>c</i>	df	SSE	<i>R</i> -Square	Rmse	AIC
1	$M_c^{\text{pred}} = ad(3) + b$	0.005	1.154	—	2	12,671	0.3708	0.4659	$-89,155$
2	$M_c^{\text{pred}} = ad(3)^c + b$	0.011	1.057	0.866	3	12,659	0.3714	0.4657	$-89,203$
3	$M_c^{\text{pred}} = a \log_{10} d(3) + b$	1.364	-1.026	—	2	13,256	0.3417	0.4766	$-86,504$

AIC, Akaike information criterion; $d(3)$, distance to the third closest station; df, degree-of-freedom; *R*-square, coefficient of determination; rmse, root mean squared error; and SSE, sum of squares due to error.



the function curves of model 2 reflect this frequency dependence. Furthermore, model 2 relies exclusively on observed M_c and $d(3)$, effectively averaging out the complexities of attenuation, including frequency dependence, velocity structure, site amplification, and basin effects. Model 2 captures the geometric dispersion and attenuation of seismic-wave amplitudes, because they propagate away from the seismic source. Figure 4 shows the predicted M_c map based on model 2 for the existing broadband seismic stations. For areas outside Chinese mainland, we employ the parameters established by Mignan *et al.* (2013) for the same geographical region. These parameters consist of $a = 5.96$, $b = -6.06$, and $c = 0.0803$. Notably, the standard deviations σ of M_c inside and outside Chinese mainland are 0.47 and 0.24, respectively. The relatively small σ value outside Chinese mainland can be attributed to the earthquake catalog's predominance in magnitudes exceeding 5, resulting in a stable estimate of M_c .

Figure 5 shows the spatial distribution of the posterior M_c estimated by the BMC technique. The distribution of the posterior M_c is more heterogeneous than that of the prior M_c , because the latter is a continuous function of the density of stations. The minimum posterior M_c value covering the entire Chinese mainland was about 3.3 (99th percentiles). The areas with posterior M_c values smaller than 1.5 and 2.0 account for 26% and 67% of the inside Chinese mainland, respectively, and are basically the same as those obtained with the prior M_c values, namely, 1.5 for 25% and 2.0 for 67% (Fig. 6). The area differences between the prior

Figure 4. Map of the predicted M_c based on prior information for the existing broadband seismic stations in Figure 1a. The standard deviations σ of M_c inside and outside Chinese mainland are 0.47 and 0.24, respectively.

and posterior M_c from $M_c = 1.0$ to 2.0 (Fig. 6) can be explained by the relatively large uncertainty ($\sigma = 0.47$) of the prior model compared with that for Switzerland ($\sigma = 0.16$), Taiwan region ($\sigma = 0.18$), South Africa ($\sigma = 0.18$), Venezuela ($\sigma = 0.19$), Greece ($\sigma = 0.29$), Iceland ($\sigma = 0.3$), and induced seismicity ($\sigma = 0.37$; see matching references in the Introduction). However, our standard deviation ($\sigma = 0.47$) is smaller than that computed by Mignan *et al.* (2013) for the Chinese catalog from 2008 to 2011, in which $\sigma = 0.56$.

M_c prediction for the planned seismic network

With the calibration of the BMC prior from the observed M_c , we can then predict the M_c distribution for any seismic network configuration, assuming that the prior model is valid when used in the same area (i.e., same seismic waveform attenuation laws). Figure 7 shows the predicted M_c distribution for the planned broadband seismic network. The results show the increase in the spatial distribution of the regions with M_c values smaller than 1.5 and 2.0. Quantitatively compared with the posterior M_c of the existing network (Fig. 5), the area with $M_c \leq 1.5$ expands by a factor of 1.5 to reach 37%, and the M_c 's of $\sim 77\%$ of inside Chinese mainland are less than 2.0. The

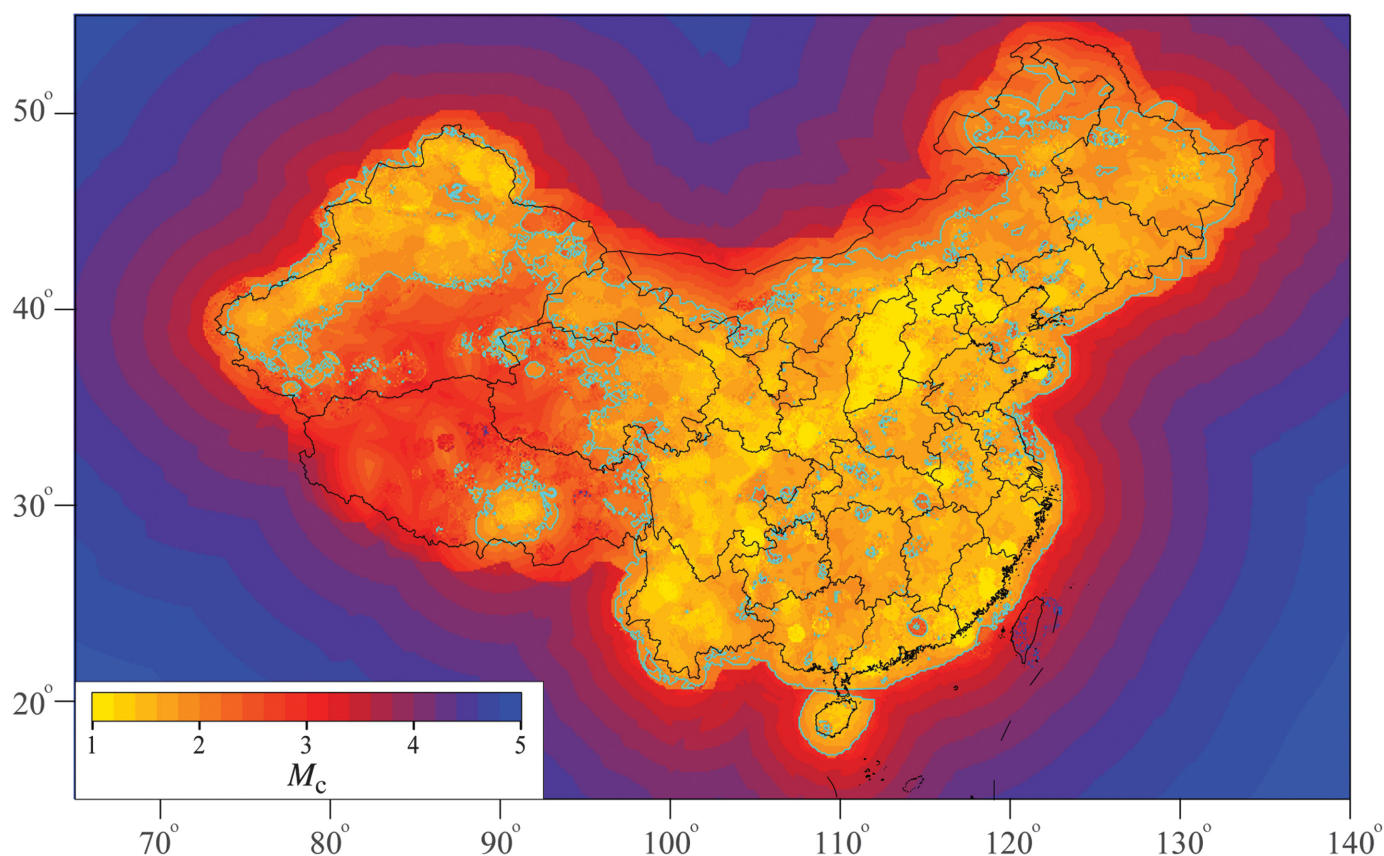


Figure 5. Map of the posterior M_c estimated by the Bayesian magnitude of completeness (BMC) method for time periods from

2009 to 2022.

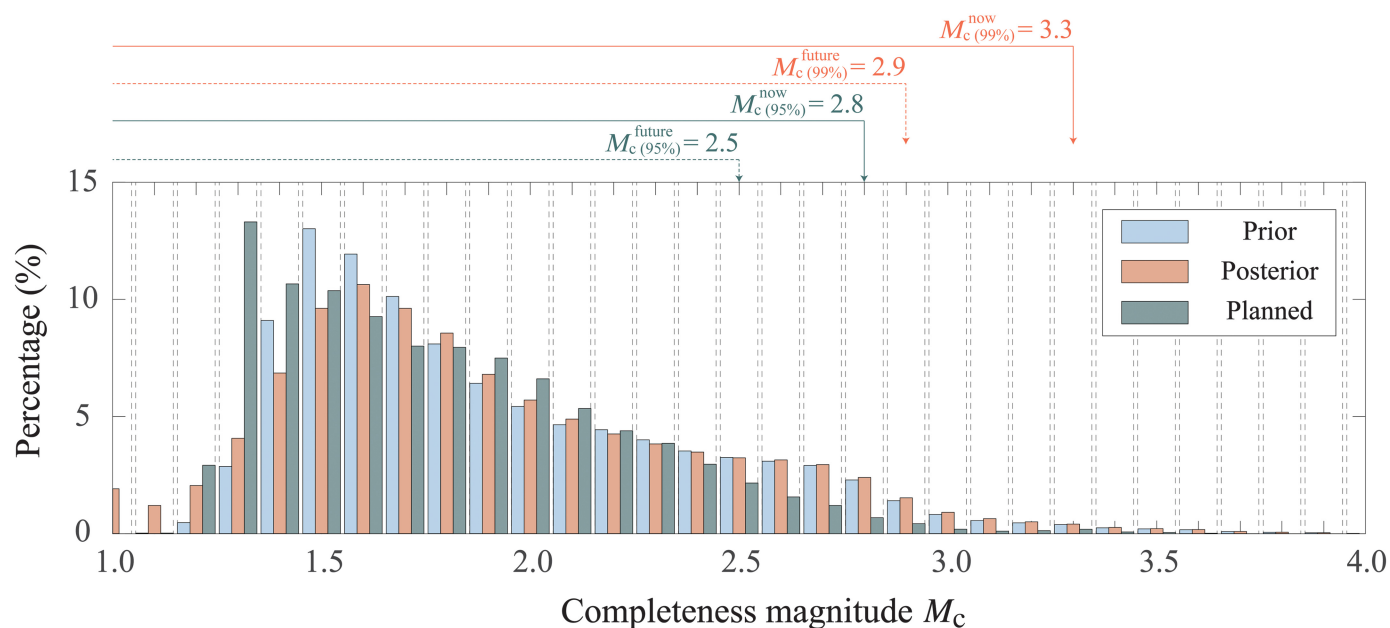
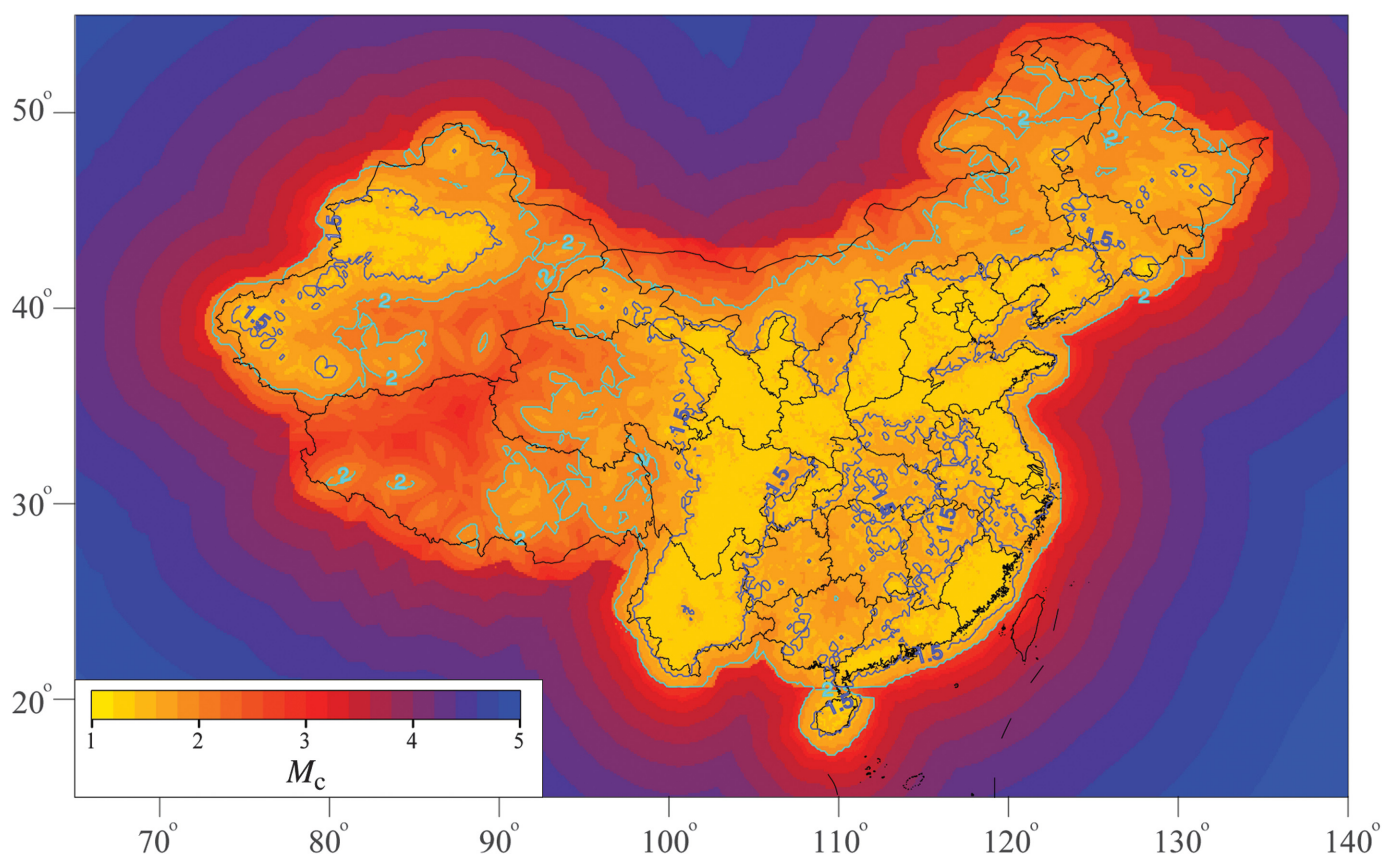


Figure 6. The binned distributions for prior, posterior, and predicted M_c values inside Chinese mainland as shown in Figures 4, 5, and 7, respectively. The top portion shows the 99th and 95th

percentiles of posterior and predicted M_c for the existing and planned networks, respectively.



monitoring capability in Xinjiang increases dramatically, followed by the central China north–south seismic belt and North China. Southeast China shows the least gain, because the existing configuration is currently sufficient with only a small improvement of monitoring capability needed.

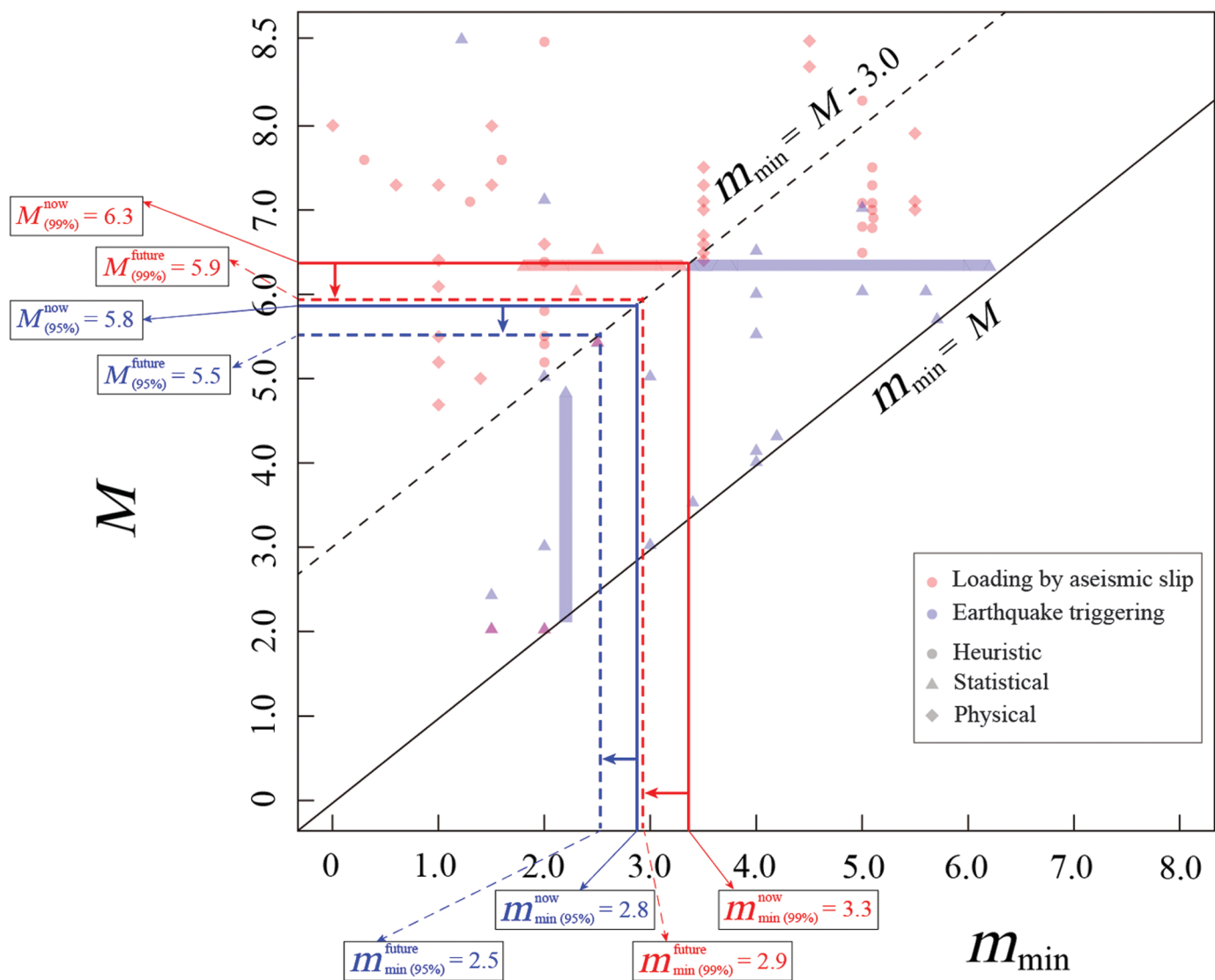
The Taiwan region, California, Switzerland, and Japan are internationally renowned for their extensive monitoring regional seismic networks. The application of the BMC method has consistently shown that M_c values in these regions generally fall below 2.0 (Mignan *et al.*, 2011; Kraft *et al.*, 2013; Tormann *et al.*, 2014). In Japan, the entire magnitude range method also estimates an M_c value of ~ 2.0 for the entire region (Nanjo *et al.*, 2010). Areas with $M_c \leq 1.5$ are primarily observed in specific high-risk urban clusters within these regions. With the implementation of the new seismic network, approximately one-fourth of Chinese mainland will achieve M_c values comparable to those in densely station-covered regions around the world. Notably, regions such as North China, the central China north–south seismic belt, the southeast coast, and the middle section of the Tianshan Mountains in Xinjiang will exhibit M_c values that rival those of the most comprehensively monitored urban clusters worldwide.

Benefits from a More Complete Catalog

Better monitoring capability leads to the detection of more small earthquakes by offering a higher resolution for active fault

Figure 7. Map of the predicted M_c based on prior information for the planned stations. The standard deviations σ of M_c inside and outside Chinese mainland are the same as Figure 4.

network structures (e.g., Ouillon and Sornette, 2011; Zhou *et al.*, 2022), clarifying the relevant components of stress that would trigger subsequent earthquakes (e.g., Mancini *et al.*, 2019), and helping to understand the earthquakes' clustering behavior (e.g., Sornette and Werner, 2005a, b) and the physics of the Earth's crust (e.g., Mignan, 2011). Mignan (2014) summarized two main hypotheses used to explain the origin of seismicity (foreshocks) observed prior to a larger earthquake (mainshock), as shown in Figure 8. The first hypothesis is named the tectonic loading process due to seismic slip (e.g., earthquake nucleation model; Das and Scholz, 1981; Ohnaka, 1992; noncritical precursory accelerating seismicity theory; Mignan, 2012), which considers the foreshock as the part of the background seismicity that is impacted by the constant loading process on the fault on which the mainshock will occur. In this view, the foreshock has significant prognostic value for the upcoming larger mainshock. The second hypothesis is known as the earthquake triggering process (e.g., the epidemic-type after-shock sequence [ETAS] model; Ogata, 1988; Helmstetter and Sornette, 2003; or self-organized criticality; Bak and Tang, 1989; Sornette and Sornette, 1989), which gives a good illustration of the triggering relations between earthquakes on each

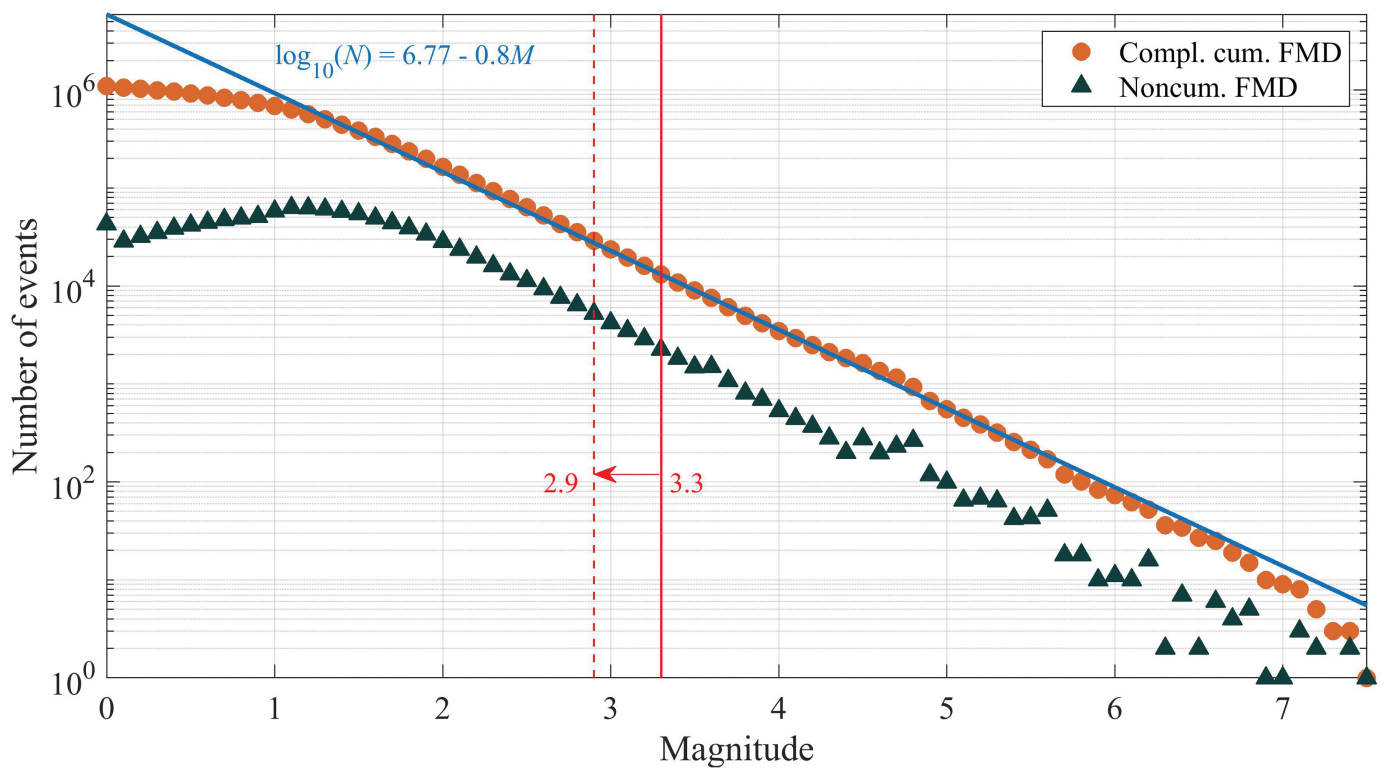


other. Indeed, according to the previous understanding in statistical seismology, as a whole, small earthquakes contribute the most in the triggering of other earthquakes, including medium-to-large earthquakes (Helmstetter, 2003; Helmstetter *et al.*, 2005). However, a current study conducted by Nandan *et al.* (2022) using augmented ETAS models shows that the earthquakes with a magnitude smaller than a certain crossover magnitude M^* are dominantly triggered by prior earthquakes with a magnitude close to the minimum magnitude M_0 at which earthquakes can trigger others (Sornette and Werner, 2005a). In contrast, events with magnitudes larger than M^* are triggered by events of magnitudes similar to their own magnitude. Considering that small earthquakes still play a significant role in earthquake prediction research, it is therefore of the utmost importance to record them as much as possible (Ebel, 2008; Mignan, 2014; Brodsky, 2019; Trugman and Ross, 2019).

The 99th percentiles of the posterior M_c for the existing network and prediction (prior) M_c for the planned network are 3.3 and 2.9, respectively (Fig. 6). Statistically, 13,101 earthquakes with $M \geq 3.3$ were observed (on average, ~ 970 per year and

Figure 8. Comparison between the existing and planned networks in Chinese mainland showing how the reduction of $m_{\min}$ allows one to observe seismic precursors for smaller mainshock magnitudes M . The emergence of abnormal foreshock behavior is observed at least at three units below the mainshock magnitude M (above the black dashed line) as expressed by equation (1). The red and blue lines depict M_c at the 99th and 95th percentiles, respectively, inside the Chinese mainland setting as $m_{\min}$, with $M_c^{99\%} = 3.3$, $M_c^{95\%} = 2.8$ for the existing network (solid lines), and $M_c^{99\%} = 2.9$, $M_c^{95\%} = 2.5$ for the planned network (dashed lines). The background comprises a total of 77 data points from the meta-analysis of 37 published studies by Mignan (2014), in which arguments are mainly based on heuristic, statistical, or physical considerations. Modified from figure 2 of Mignan (2014).

80 per month) from January 2009 to June 2022. According to the Gutenberg–Richter law that fits the empirical magnitude distribution for magnitudes larger than 3.3 (a -value of 6.77 and b -value of 0.80; Fig. 9), the available seismicity samples ($M \geq 2.9$) will increase by a factor of at least 2 to reach $\sim 27,500$. We predict that the complete catalog of the planned broadband seismic



network will record ~ 2040 and 170 earthquakes per year and month, respectively, which is at least a twofold increase compared with the number of earthquakes recorded by the existing network. The improvement would potentially strengthen both predictive skills of the statistics- and physics-based models because the sample number of earthquakes would be increased to better capture the relevant physical processes underlying the nucleation or triggering of large earthquakes.

Equation (1) that derives from a meta-analysis of 37 foreshock studies published from 1982 to 2013 shows that seismic precursors are most likely to be observed at least at three units below the mainshock magnitude M (Fig. 8). Figure 8 modified from figure 2 of Mignan (2014) uses a total of 77 data points (solid circles, triangles, and diamonds for heuristic, statistical, or physical considerations, respectively) from literature surveys as background. The 77 data points are classified into (1) loading processes caused by aseismic slip (e.g., the emergence of an abnormal foreshock behavior) represented in red and (2) earthquake triggering process (e.g., the normal behavior of seismicity) represented in blue, respectively. The emergence of an abnormal foreshock behavior is observed approximately in the regime represented by equation (1), namely above the dashed line in Figure 8. Mignan (2014), therefore, concluded that earthquake prediction research might be fostered in the future via increased statistical robustness obtained from analyses of large catalog containing microseismicity. This could exploit the regime $m_{\min} < M - 3.0$ indicated by equation (1), for which anomalous foreshocks seem to emerge.

Figures 6 shows that the M_c at the 99th percentiles inside Chinese mainland is 3.3 for the existing network and 2.9 for

Figure 9. Complementary cumulative and density frequency-magnitude distribution (FMD) for earthquakes inside Chinese mainland from 2009 and 2022. The vertical solid and red dashed lines represent the 99th percentile M_c values for the existing and planned networks, respectively. The blue line depicts the Gutenberg-Richter law fit for earthquakes with magnitudes larger than 3.3.

the new network, respectively. Considering the optimal use of data, namely $m_{\min} = M_c$, we find that the magnitude M of mainshocks prior to which anomalous foreshock seismicity most likely to be found to emerge is decreased from 6.3 for the existing network to 5.9 for the future network (Fig. 8). In addition, M is decreased from 5.8 for the existing network to 5.5 for the future network if taking M_c values from the 95th percentiles (Figs. 6, 8). Assuming that earthquakes in Chinese mainland are potentially destructive for magnitude M larger than 5, the future seismic network will achieve the goal of a 77% coverage of inside Chinese mainland for such optimal seismic precursor-based earthquake prediction research ($m_{\min} = 2.0$). For the same goal, the existing seismic network has only a 67% coverage.

Conclusions and Discussion

The performance of the broadband seismic network will significantly improve by simply densifying the existing layout. For more than half of Chinese mainland, the interstation distance will soon be smaller than 100 km (and smaller than 50 km and 25 km for 27% and 6% of the mainland, respectively). The area with an interstation distance smaller than 50 km and 25 km that are to be covered by the new network

are expanded by factors 2 and 8, respectively. In this study, we quantified the higher-resolution detection of the frequent smaller earthquakes inside Chinese mainland via the completeness magnitude (M_c) metric for the existing and planned seismic networks. Using the best-performance prior model ($\sigma = 0.47$) of the BMC method calibrated on the Chinese catalog from 1 January 2009 to 26 June 2022, we predicted the spatial distribution of M_c for the future network based on its network configuration. Compared with the posterior M_c of the existing network, the area with $M_c \leq 1.5$ that is covered by the new network expands by a factor of 1.5 to reach 37%, and the M_c 's of $\sim 77\%$ inside Chinese mainland are smaller than 2.0. If 99% of Chinese mainland is at present covered down to $M_c = 3.3$, this value will soon fall to $M_c = 2.9$. We predicted that the complete catalog of the planned network will record a factor of 2 more earthquakes per year and month compared with that of the existing network (for $a = 6.77$ and $b = 0.80$ in the Gutenberg–Richter law $\log_{10} N = a - bM$, in which N represents the number of events of magnitude larger than or equal to M and $M \geq M_c$). Taking $m_{\min}$ inside Chinese mainland as $M_c = 2.0$ for the future network and assuming earthquakes to be potentially damaging at $M \geq 5$, the new network shall achieve the goal of almost total coverage for optimal seismic-based earthquake prediction research in 77% inside Chinese mainland. Our investigation provides a useful reference for the real functioning and further optimization of the new networks in Chinese mainland.

Improved monitoring capability, enabling the detection of more small earthquakes, offers several potential benefits. It provides higher resolution for active fault network structures, helps clarify the relevant stress components triggering subsequent earthquakes, and enhances our understanding of earthquake clustering behavior and Earth's crust physics. In addition, better monitoring capability plays a crucial role in practical hazard risk analysis and prediction. For example, it significantly contributes to Seismic Hazard Assessment (SHA; Cornell, 1968; Reiter, 1990; Panza *et al.*, 2001, 2022) by facilitating the construction of large earthquake scenarios with extended rupture. By utilizing a more comprehensive catalog that includes smaller seismicity, fault ruptures for large earthquakes can be characterized with higher precision and resolution, resulting in more precise and reliable SHA outcomes.

The improvement in computing power makes the adoption of artificial intelligence (AI) in the routine monitoring of seismic networks possible. Previous tests showed that more small earthquakes by at least a factor of 10 could be detected by AI compared with that produced by the current processing techniques (e.g., Beroza *et al.*, 2021). Therefore, we can expect that the monitoring capabilities of the seismic network will usher a comprehensive and large improvement in the near future with the revolution in seismic observational technology enabled by AI and the further upgradation of the software in the near future.

Data and Resources

The earthquake catalog and information of the existing and the planned seismic networks were provided by China Earthquake Networks Center (CENC; doi: [10.11998/SeisDmc/SN](https://doi.org/10.11998/SeisDmc/SN)) through the internal link (Earthquake Cataloging System at China Earthquake Administration [CEA] available at <http://10.5.160.18/console/index.action>, last accessed March 2022). The Bayesian magnitude of completeness (BMC) method used in the study is from Mignan *et al.* (2011) and can also be found in the R package rseismNet at <https://github.com/amignan/rseismNet> (last accessed September 2023). The total investment information about the National System for Fast Report of Intensities and Earthquake Early Warning Project of China is from the CENC (<https://www.cenc.ac.cn/cenc/zt/361404/361414/361563/index.html>, last accessed January 2022). The application of BMC in Venezuela can be seen in the online report entitled “Bayesian estimation of the spatial variation of the completeness magnitude for the Venezuelan seismic catalogue” by Vásquez, R., and L. Bravo de Guenni (<https://www.statistics.gov.hk/wsc/CPS204-P47-S.pdf>, last accessed September 2021). The supplemental material for this article includes visualizations that illustrate the distribution of earthquakes across the Chinese mainland, comprehensive maps that delineate interstation distances for both the existing and planned networks, as well as the standard deviation of observed M_c values.

Declaration of Competing Interests

The authors acknowledge that there are no conflicts of interest recorded.

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